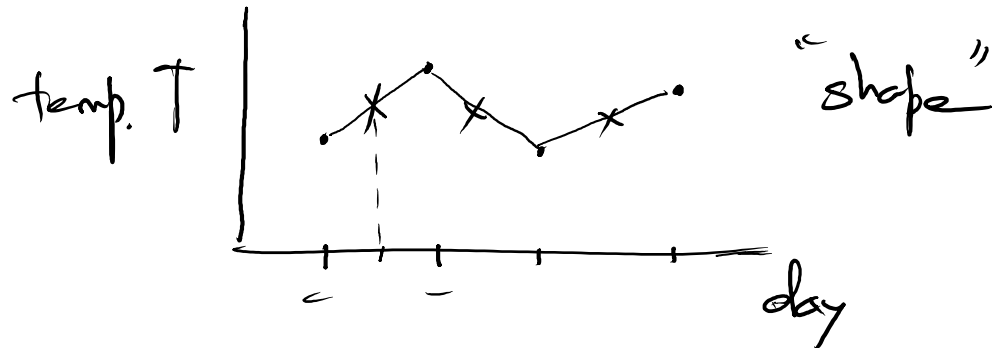
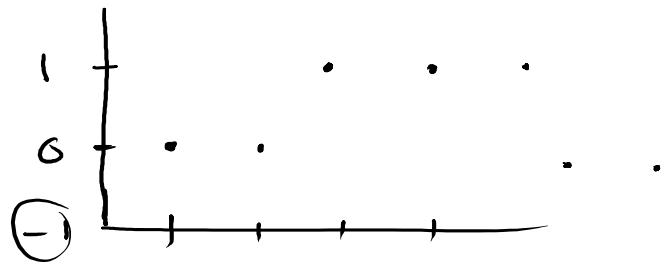


① Max daily temperature each day. (DTCS)



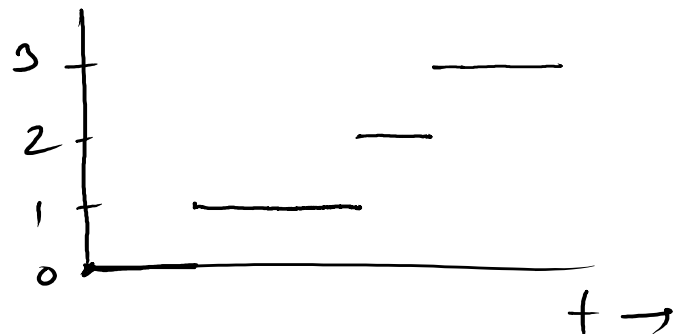
② rain or not every day (DTDS)

$$S \in \{0, 1\}, J \in \{0, 1, 2, \dots\}$$



③ # accidents. (CTDS)

$$S \in \{0, 1, 2, \dots\}, J \in \mathbb{R}$$



$$P(\underline{A}) = P(A|B)$$

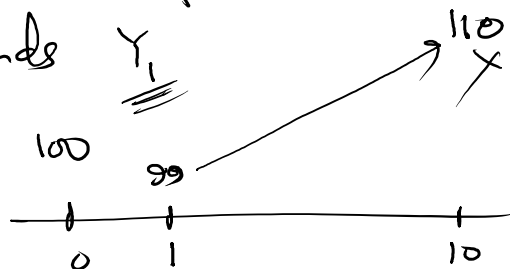
$$Y_0 = 100$$

$$\rightarrow P[\underline{Y_{10}} = 110] = p^{10} > 0$$

$$\rightarrow P[\underline{Y_{10}} = 110 \mid \underline{Y_1 = 99}] = \underline{0}$$

$P(Y_{10})$ depends

Y_1



$$\begin{array}{cc} \checkmark +1 & -1 \\ \checkmark -1 & +1 \end{array}$$

$$\underline{Y_+}$$

$$\underline{Y_1} = \begin{cases} 101, p \\ 99, 1-p \end{cases} \quad \checkmark$$

$$\underline{Y_2} = \begin{cases} 102, p^2 \\ 100, 2p(1-p) \\ 98, (1-p)^2 \end{cases} \quad \checkmark$$

(1) X_t ?? $X_t = \begin{cases} 1, & p \\ -1, & 1-p \end{cases}, \quad p=0.5$

$$E[X_t] = 1 \times p + (-1) \times (1-p) = p + p - 1 = 2p - 1 = \underline{\underline{0}}$$

$$V[X_t] = (1-0)^2 \times 0.5 + (-1-0)^2 \times 0.5 = 0.5 + 0.5 = \underline{\underline{1}}$$

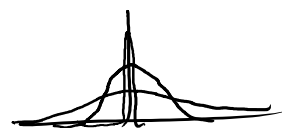
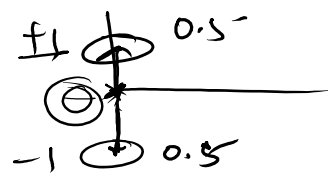
(2) Y_t ??

$$\text{Cov}(X_i, X_j) = 0$$

$$Y_t = \underline{Y_0} + \underline{X_1} + \underline{X_2} + \dots + \underline{X_t}$$

$$\rightarrow \underline{E[Y_t]} = Y_0 + 0 + 0 + \dots + 0 = \underline{\underline{Y_0}}$$

$$\underline{V[Y_t]} = 0 + 1 + 1 + \dots + 1 = \underline{\underline{t}}$$



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moments

Saturday, May 24, 2025 9:51 AM

$$\textcircled{1} \text{Cov}(\underline{aX + bY}, \underline{cW + d}) = ac \text{Cov}(X, W) + 0 + bc \text{Cov}(Y, W) + 0$$

$$\textcircled{2} \text{Cov}(Y_s, Y_t)$$

assume $s < t$

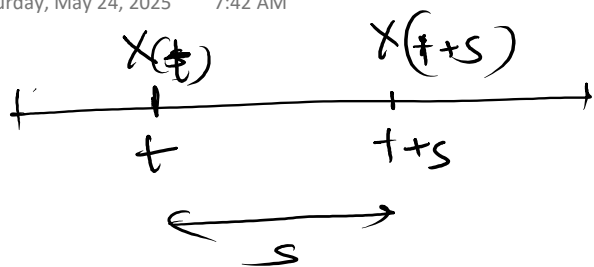
$$= \text{Cov}(Y_s, Y_s + X_{s+1} + X_{s+2} + \dots + X_t)$$

$$= \underline{s}$$

$$\text{Cov}(Y_s, Y_t) = \min(s, t)$$

$$\textcircled{3} X_t = X_{t-1} + I_t, \quad I_t = \begin{cases} +1, & e^{-|X_{t-1}|} \\ -1, & 1 - e^{-|X_{t-1}|} \end{cases}$$

$$X_t = \begin{cases} X_{t-1} + 1, & e^{-|X_{t-1}|} \\ X_{t-1} - 1, & 1 - e^{-|X_{t-1}|} \end{cases} \quad \underline{\text{Markov}}$$



$$\underline{X(t)} \sim \text{Poisson}(\lambda t)$$

$$\begin{aligned} \text{Cov}(X(t), X(\underline{t+s})) &= \text{Cov}(X(t), \overbrace{X(t) + X(t+s) - X(t)}^{\text{independent}}) \\ &= \text{Var}(X(t)) + 0 \\ &= \lambda t \end{aligned}$$

$$\left[\binom{n}{x} \begin{matrix} \xrightarrow{x \text{ (+1)}} \\ \xleftarrow{n-x \text{ (-1)}} \end{matrix} \right] \rightarrow \underline{\underline{y}} = \frac{x \times 1 + (n-x) \times (-1)}{2x - n}$$

$$P(Y_n = y) = ?$$

$$x = \binom{\frac{n+y}{2}}{\uparrow} \text{ (up)}$$

$$n - \left(\frac{n+y}{2}\right) = \frac{n-y}{2} \text{ (down)}$$

$$\binom{n}{x} p^x \cdot (1-p)^{n-x} \rightarrow P(Y = \underline{\underline{2x-n}})$$

$$P[\underline{\underline{Y}}_n = \underline{\underline{y}}] = \binom{n}{\frac{n+y}{2}} p^{\frac{n+y}{2}} \cdot (1-p)^{\frac{n-y}{2}}$$

①

$\rightarrow \underline{\underline{\frac{n+y}{2}}} \rightarrow \underline{\underline{even}}$

